Dynamic Performance of Brushless DC Motors With Unbalanced Hall Sensors

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***Abstract*—Brushless dc (BLDC) motors controlled by Hall-effect sensors are widely used in various applications and have been extensively researched in the literature, mainly under the assumption that the Hall sensors are ideally placed 120 electrical degrees apart. However, this assumption is not always valid; in fact, sensor placement may be significantly inaccurate, especially in medium- and low-precision BLDC machines. This paper shows that misplaced Hall sensors lead to unbalanced operation of the inverter and motor phases, which increases the low-frequency harmonics in torque ripple and degrades the overall drive performance. The paper also presents several average-filtering techniques that can be applied to the original Hall-sensor signals to mitigate the effect of unbalanced placement during steady-state and transient operations. The proposed methodology is demonstrated by modeling and hardware, and is shown to achieve dynamic performance similar to that of a BLDC motor with accurately positioned Hall sensors.**

***Index Terms*—Average filtering, brushless dc (BLDC) motor, extrapolation, Hall-effect devices.**

I. Introduction

BRUSHLESS dc (BLDC) motors are often considered in various electromechanical applications and generally have been investigated quite well in the literature [1]–[6]. The techniques used to control the inverter transistors can be placed into two major categories: those that require Hall sensors [1], [5], [6] and those that are based on a sensorless approach, for example, that use back electromotive force (EMF) zero-crossings [6]. An advantage of the first approach is its relatively simple implementation and reliable operation with variable mechanical loads, even at very low speeds (where sensorless control may not always be effective).

The theoryand modeling of BLDC motors driven by Hall sensors have been developed by many researchers under one common assumption—that the Hall sensors are placed exactly 120 electrical degrees apart. However, in many low-cost machines, this assumption may not hold true, and the distribution of relative displacements may, in fact, be quite significant. An example of Hall-sensor placement in the typical industrial motor considered in this paper is illustrated in Fig. 1. As can be seen, the Hall sensors (H1, H2, and H3) are mounted on a printed circuit (PC) board placed outside the motor case and react to the magnetic field produced by a permanent magnet tablet attached to the rear end of the motor's shaft. In an ideal case, the axes of the sensors should be 120 degrees apart, which, in practice, is difficult to achieve with high accuracy. Moreover, the errors in positioning of the sensors may be different for different phases. The dashed axes in Fig. 1 correspond to the desired positioning of the sensors, and the solid lines denote their actual positions. As was determined, the absolute error of sensor placement may reach several mechanical degrees, which translates into an even greater error in electrical degrees for machines with a high number of magnetic poles.

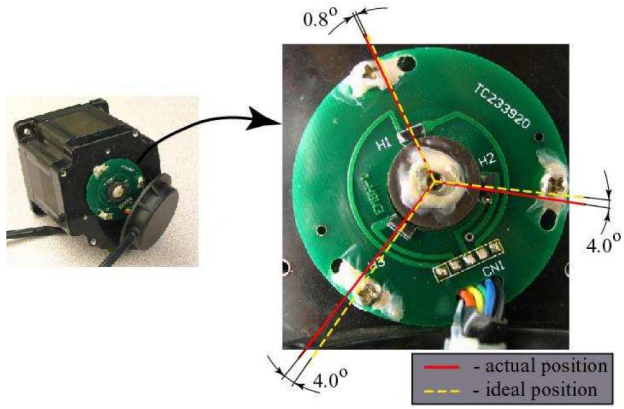


Fig. 1. Hall-effect sensor placement in a typical BLDC motor.

Although other configurations and/or mounting of Hall sensors are also possible in different BLDC machines, the effect of their misalignment leads to similar consequences. In general, insufficiently precise positioning of the Hall sensors causes unbalanced operation of the motor inverter, with some phase(s) conducting for longer and other phase(s) conducting for shorter time intervals. The resulting unbalance among the phases leads to a number of adverse phenomena, such as an increase in torque pulsation, vibrations, and acoustic noise, as well as reduced overall electromechanical performance.

Although there exists a large number of publications on BLDC drives, after conducting extensive literature search, we have found that only few address the unbalanced Hall sensors. A misalignment of Hall sensors was documented in [7], where the authors investigated a relatively sophisticated (expensive) BLDC motor drive with an advanced observer-based torque

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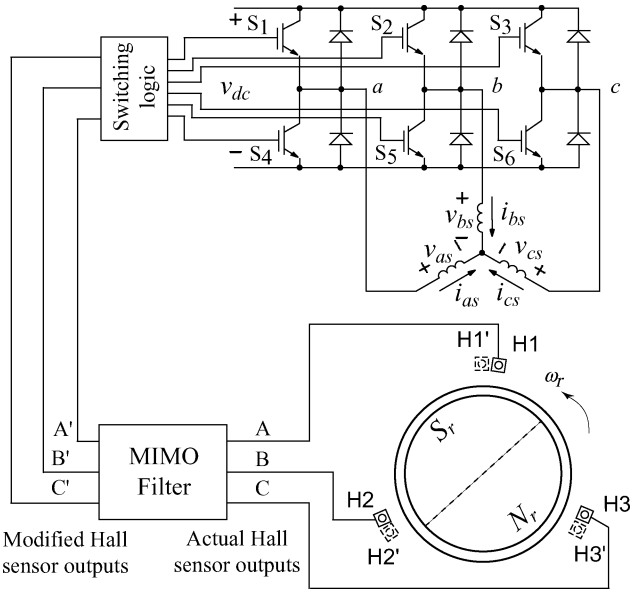


Fig. 2. Brushless dc motor drive system with filtering of Hall-sensor signals.

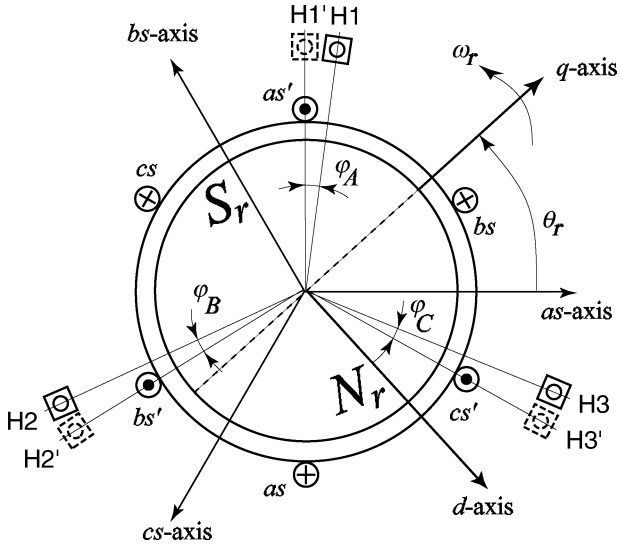


Fig. 3. PMSM with unbalanced Hall sensors.

ripple mitigation control. For small-scale machines, the accuracy of Hall-sensor positioning may also be a problem. The authors of [8] demonstrate their carefully made prototype and relate the unsymmetrical phase currents to the Hall-sensor inaccuracy. Although the position errors due to low-precision Hall sensors appears to be a known problem, our extensive literature search has not yield many solutions. An approach of manually realigning the sensors requires opening the machine (or its back side) and adjusting the sensors until the required accuracy is achieved, which is not very practical especially for large quantity of motors. Introducing additional hardware circuitry is also not desirable because it leads to increased complexity and cost of the drive. The operation of a low-precision BLDC motor with misaligned Hall sensors was described in [9], where a simple averaging of the time intervals was proposed to improve the steady-state operation. This work was further developed to consider the motor operation during the transients in [10].

As low-cost/low-precision BLDC motors are now becoming widely available and used in a variety of applications, the misalignment of Hall sensors requires more detailed attention. This paper focuses on a typical three-phase BLDC motor-inverter system, as shown in Fig. 2. We present a filtering methodology that can be applied directly to the original Hall-sensor signals to produce a modified set of signals that is used to drive the inverter depicted in Fig. 2. The present manuscript extends the work reported by the authors in [10] and makes the following overall contributions.

1) The paper describes the phenomenon of nonideal placement of Hall sensors based on a hardware prototype and a detailed switching model.
2) We propose a simple but very effective and practical filtering technique to improve the overall performance of a BLDC motor-drive system with significant unbalance in Hall-sensor positioning.
3) This paper generalizes the approach of filtering the Hall-sensor signals presented in [10] and provides the experimental results. We show that the performance of the BLDC motor with the proposed filters approaches that of a motor with ideally placed Hall sensors.
4) The proposed methodology does not require any additional and/or special circuitry or hardware. Our solution can be implemented (programmed) with a basic (possibly already existing) motor controller, and therefore, may be useful for many applications.

II. Permanent Magnet BLDC Machine Model

A. Detailed Model

To analyze the impact of unbalanced Hall sensors on BLDC motor performance, a permanent-magnet synchronous machine (PMSM), shown in Fig. 3, is considered here. In Fig. 3, $\mathrm{H}\{1,2,3\}$ and $\mathrm{H}\{1,2,3\}'$ denote the actual and ideal axes (positions) of the Hall sensors, respectively; and φ_A, φ_B, and φ_C denote the absolute errors in sensor placement in electrical degrees. Based on commonly used assumptions, the stator voltage equation may be expressed as follows [1]–[4]:

$$\mathbf{v}_{abcs} = \mathbf{r}_s \mathbf{i}_{abcs} + \frac{d\boldsymbol{\lambda}_{abcs}}{dt} \tag{1}$$

where $\mathbf{f}_{abcs} = [f_{as} \quad f_{bs} \quad f_{cs}]^T$, and $\mathbf{f}$ may represent the voltage, current, or flux linkage vectors. Also, $\mathbf{r}_s$ represents the stator resistance matrix. In the case of a motor with nonsinusoidal back EMF, the back EMF is assumed to be half-wave symmetric and contain spatial harmonics. Therefore, the stator flux linkages and electromagnetic torque may be written as [4]

$$\boldsymbol{\lambda}_{abcs} = L_s \mathbf{i}_{abcs} + \lambda'_m \sum_{n=1}^{\infty} K_{2n-1} \begin{bmatrix} \sin((2n-1)\theta_r) \\ \sin\left((2n-1)\left(\theta_r - \frac{2\pi}{3}\right)\right) \\ \sin\left((2n-1)\left(\theta_r + \frac{2\pi}{3}\right)\right) \end{bmatrix} \tag{2}$$

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$$T_e = \frac{3P}{4}\lambda'_m \sum_{n=1}^{\infty} K_{2n-1} \begin{bmatrix} i_{as} \\ i_{bs} \\ i_{cs} \end{bmatrix}^T \begin{bmatrix} \cos((2n-1)\theta_r) \\ \cos\left((2n-1)\left(\theta_r - \frac{2\pi}{3}\right)\right) \\ \cos\left((2n-1)\left(\theta_r + \frac{2\pi}{3}\right)\right) \end{bmatrix} \quad (3)$$

where L_s is the stator-phase self-inductance and λ'_m is the magnitude of the fundamental component of the phase modulation (PM) magnet flux linkage. The coefficients K_n denote the normalized magnitudes of the nth flux harmonic relative to the fundamental, i.e., $K_1 = 1$.

A detailed model of the system shown in Fig. 2 was developed and implemented in Matlab Simulink [11] using the toolbox [12]. The 120° inverter logic was implemented according to the standard table [3], [5], [6].

B. Model Verification

To study the phenomena of unbalanced Hall-effect sensors, we tested a batch of industrial BLDC motors for possible variation in the severity of sensor unbalance parameters among the samples. The parameters of the motor used in the verification studies presented in this paper are summarized in the Appendix. Since it is difficult to precisely measure the positioning errors from the physical mounting of Hall sensors (see Fig. 1), these errors can be measured very accurately indirectly by measuring the phase difference between the actual back EMFs and the Hall-sensor output signals. Thus, for the given motor, the actual back EMFs and the Hall-sensor output signals were captured experimentally from which the absolute sensor positioning errors were determined to be $+0.8°$, $-4°$, and $-4°$ mechanical degrees for phases A, B, and C, respectively. Although some other motors had better or worse precision, the considered sample was assumed to be sufficiently representative. The measured back EMF waveforms for this motor have been included in [10], and are not repeated here due to space limitations. To improve the accuracy of the model, the spatial harmonics were included according to (2) and (3). The most significant harmonic coefficients are summarized in the Appendix. If desired, additional coefficients could be considered for the detailed model; however, higher order harmonics were found to be less significant.

To study the effect of unbalanced Hall sensors in steady state, a number of experiments were carried out using several commercially available BLDC Hall-sensor-based drivers (Maxon EC Amplifier DEC 50 and Anaheim Automation MDC 150-050) as well as our own prototype driver (see Section V), all producing the same results. Without the loss of generality, an operating point determined by a mechanical load of 0.9 N·m is included here. For this study, the motor inverter was supplied with $V_{\rm dc} = 40$ V, resulting in a speed of 2458 r/min under the given mechanical load. The measured phase currents were captured and are shown in Fig. 4 (top).

The simulated phase currents for the same steady-state operating condition are shown in Fig. 4 (middle). As can be seen in Fig. 4, the detailed model predicts the phase currents very closely and agrees with the measured waveforms. This study confirms the accuracy of the developed detailed model. As can be observed in Fig. 4 (top and middle), the motor phases are energized for unequal periods of time, and the currents are asymmetrically distorted. Although a given operating point is slightly above the machine's ratings, the results obtained at other loading conditions were distorted in a similar way. In general, this very phenomenon does not depend on the loading and/or operating condition of the motor, and it will be present as long as the inverter transistors are commutated incorrectly. For comparison, the machine operation with ideally placed Hall sensors was also simulated, and the resulting phase currents are plotted in Fig. 4 (bottom). As can be seen from the figure, the conduction intervals and current waveforms should be balanced among the phases.

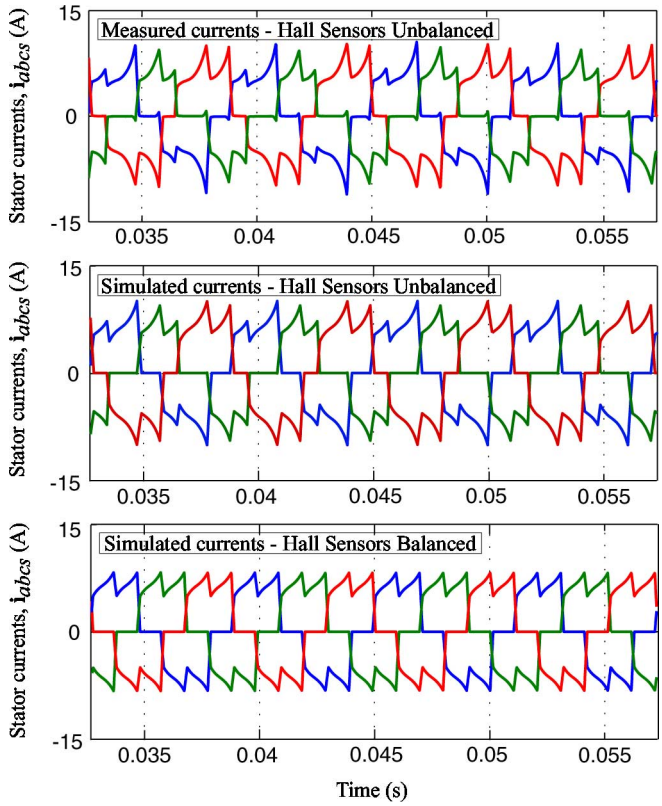


Fig. 4. Measured and simulated phase currents.

The asymmetrical stator currents also distort the developed electromagnetic torque. Since it is hard to measure the actual instantaneous electromagnetic torque in practice, the torque waveforms were predicted using detailed simulations for the two cases: 1) ideal case—the Hall sensors are precisely placed, with zero errors and 2) the actual case—the Hall sensors are placed with errors equal to those of the sample motor. The predicted torque waveforms are shown in Fig. 5, and the corresponding harmonic spectra are depicted in Fig. 6, wherein a significant difference can be observed. As can be seen in Figs. 5 and 6 (ideal case, top), the torque waveform contains very strong harmonics at the frequency of 984 Hz, which corresponds to the six-pulse inverter operation at the given motor speed. This harmonic is

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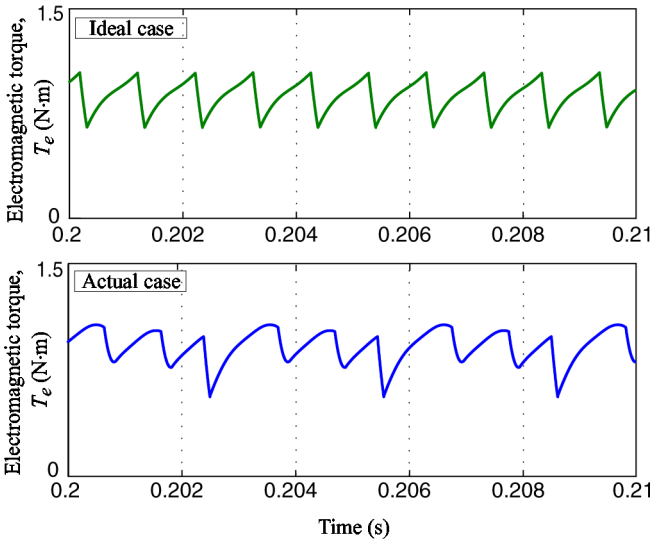


Fig. 5. Electromagnetic torque waveforms.

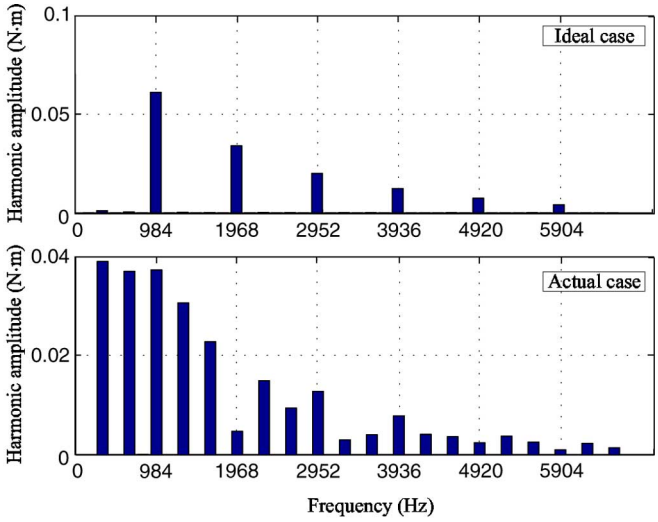


Fig. 6. Electromagnetic torque harmonic content.

expected to dominate under normal operation. However, the torque corresponding to the actual case (see Figs. 5 and 6, bottom) has a much richer spectrum, with two very strong harmonics below 984 Hz. These lower harmonics are particularly undesirable as they result in increased mechanical vibration and acoustic noise.

Another way to qualitatively compare these waveforms is to evaluate their total harmonic distortion (THD). In the literature, the THD is commonly defined in terms of harmonic content of the waveform related to its fundamental component or its rms value [13]. Since both definitions are related to each other, without the loss of generality, here we evaluate the THD of the torque waveforms with respect to the 984 Hz, which should be the dominant component under normal operation. Thus, the calculated THDs are 69.7% and 189.6% for the ideal and actual case, respectively. This also shows a significant increase in distortion.

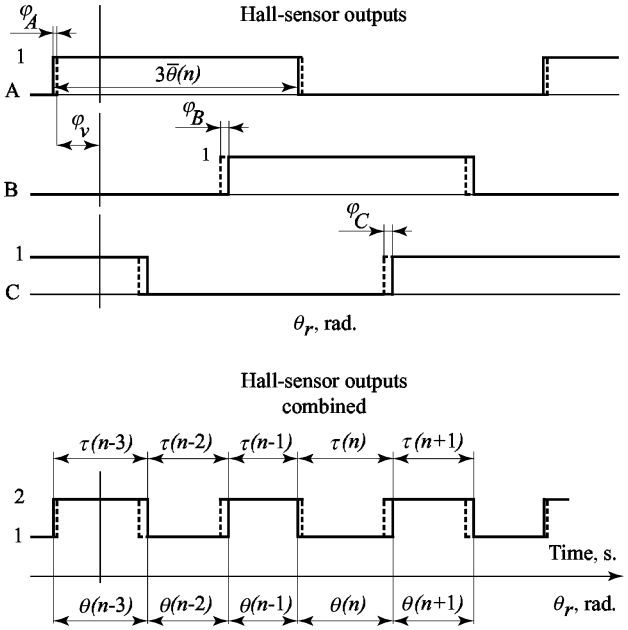


Fig. 7. Ideal and actual Hall-sensor output signals.

The detailed analysis of vibrations and acoustic signatures of BLDC machines is very important [14]–[16], and, in general, requires information about the machine's design and possible electromechanical resonances that is beyond the scope of this paper. This paper focuses instead on establishing a methodology by which the BLDC motor operation can be simply restored as close as possible to the ideal case of balanced phase currents, depicted in Fig. 4 (bottom), resulting in improved electromagnetic torque [shown in Figs. 5 and 6 (top)].

III. Filtering Hall Signals

To better understand how to correct the Hall-sensor signals, it is instructive to consider the diagram depicted in Fig. 7, where each Hall sensor is assumed to output a logical signal (0 or 1) of 180 electrical degrees. Here, the angle φ_v denotes a possible delay or advance in firing [1], and φ_A, φ_B, and φ_C are the respective sensor-positioning errors in each phase. When the ideal motor is running, the Hall sensors produce square wave signals displaced by exactly 120 electrical degrees relative to each other (see Fig. 7, dashed line). Combining (adding) all three ideal outputs produces a square wave (see Fig. 7 bottom, dashed line) with a period equal to one-third of a Hall-sensor period, which is equal to 60 electrical degrees.

When the sensors are shifted from their ideal positions (see Fig. 7, solid line), the resulting combined waveform becomes distorted, resulting in nonuniform angular intervals $\theta(n)$ between two successive switching events. The durations of intervals $\theta(n)$ are denoted here by $\tau(n)$. As can be observed in Fig. 7, the rising edge of interval $\theta\,(n-3)$ and the falling edge of interval $\theta\,(n-1)$ correspond to switching of the same sensor (in this case, the sensor of phase A). Therefore, the following

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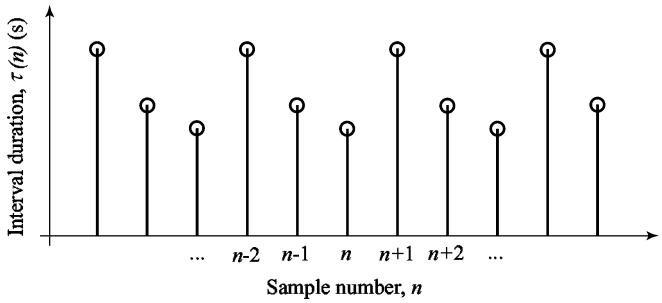


Fig. 8. Sequence of time intervals $\tau(n)$ for unbalanced Hall sensors.

holds true

$$\bar{\theta}(n) = \frac{1}{3}\left[\theta(n-3) + \theta(n-2) + \theta(n-1)\right] \tag{4}$$

which is the average angle between two ideal successive switching events, and is equal to $\pi/3$.

This paper presents a methodology to approximate the ideal Hall signals corresponding to $\mathrm{H}\{1,2,3\}'$ by appropriately modifying (filtering) the signals from actual sensors $\mathrm{H}\{1,2,3\}$. The proposed method works by finding an interval duration $\bar{\tau}(n)$ corresponding to $\bar{\theta}(n)$ by means of averaging and/or extrapolating the time intervals $\tau(n)$. Once $\bar{\tau}(n)$ is known, it is used for estimating the correct timings for commutating inverter transistors.

For clarity, the sequence $\tau(n)$ (see Fig. 7, bottom) is reproduced in Fig. 8 as a discrete-time signal with period $N = 3$, wherein the samples are the actual values of $\tau(n)$. Clearly, the nonuniform values of $\tau(n)$ cause undesirable harmonics in phase currents and torque waveforms. The frequency content of $\tau(n)$ can be evaluated by using the discrete-time Fourier series (DTFS) [17], so that the signal can be written as

$$\tau(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N} \tag{5}$$

where the Fourier coefficients $\{c_k\}$, $k = 0, 1, \ldots, N-1$, provide the description of $\tau(n)$ in the frequency domain. In our case, the signal $\tau(n)$ has one zero-frequency component and two components with frequencies of $2\pi/3$ and $4\pi/3$ radians per sample; these two frequencies should be filtered out.

In this paper, we present a methodology for removing the undesirable harmonics based on filtering the original Hall-sensor signals. Moreover, to simplify the problem of designing the required multi-input multi-output (MIMO) filter (see Fig. 2), we propose applying the filtering directly to the sequence $\tau(n)$ (see Fig. 8), which internally reduces the problem to the single-input single-output (SISO) filter. Therefore, it is necessary to filter out the undesirable harmonics in $\tau(n)$. An appropriate filter may be constructed using the following general formula

$$\bar{\tau}(n) = \sum_{m=1}^{M} b_m \tau(n-m) \tag{6}$$

where M is the order of the filter corresponding to the number of previous points taken into account and b_m are the weighting coefficients that depend on a particular filter realization and its numerical property. Without the loss of generality, in this paper, we propose two families of suitable filters: 1) basic averaging filters and 2) extrapolating filters, whereas other filters may also be derived based on (6).

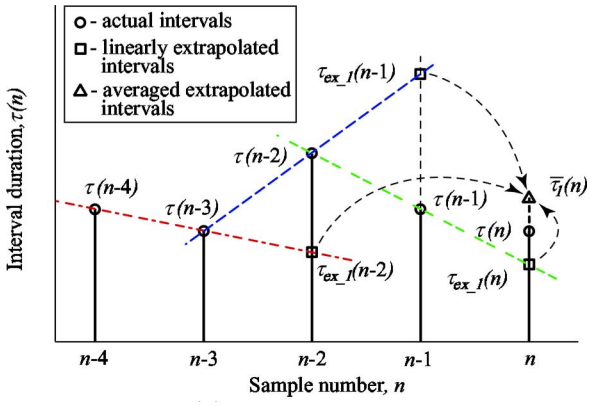


Fig. 9. Computing $\bar{\tau}_l(n)$ using linear extrapolation and subsequent averaging.

A. Basic Averaging Filters

In this approach, the coefficients in (6) can be defined as

$$b_m = \frac{1}{M}. \tag{7}$$

With this implementation, the six- and three-step filters can be represented, respectively, as

$$\bar{\tau}_{a6}(n) = \frac{1}{6}\sum_{m=1}^{6} \tau(n-m) \tag{8}$$

and

$$\bar{\tau}_{a3}(n) = \frac{1}{3}\sum_{m=1}^{3} \tau(n-m). \tag{9}$$

Here, the subscript "a" denotes this basic averaging procedure. The order of the filter should be selected with care considering that the undesirable harmonics, in this case, $2\pi/3$ and $4\pi/3$ should be suppressed.

B. Extrapolating Filters

When the drive system experiences a speed transient, such that $\tau(n)$ may no longer be periodic, it may be advantageous to consider an extrapolation (prediction) of samples $\tau(n)$ to better cope with the acceleration and deceleration of the motor. Let us first consider a linear extrapolation approach as depicted in Fig. 9. Here, each subsequent step $\tau_{ex_l}(n)$ is linearly extrapolated based on a two-step history, as follows:

$$\tau_{ex_l}(n) = 2\tau(n-1) - \tau(n-2). \tag{10}$$

To ensure the cancellation of undesirable harmonics, the values $\tau_{ex_l}(n)$ are then averaged to yield an analogue to $\bar{\tau}_{a3}(n)$ in (9), as follows:

$$\bar{\tau}_l(n) = \frac{1}{3}\left[\tau_{ex_l}(n) + \tau_{ex_l}(n-1) + \tau_{ex_l}(n-2)\right]. \tag{11}$$

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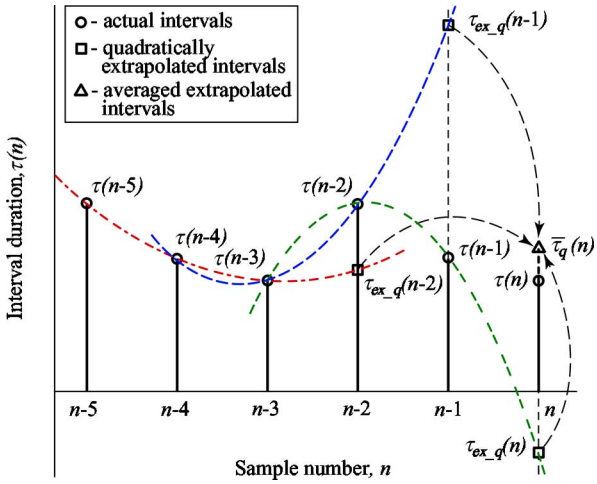


Fig. 10. Computing $\bar{\tau}_q(n)$ using quadratic extrapolation and subsequent averaging.

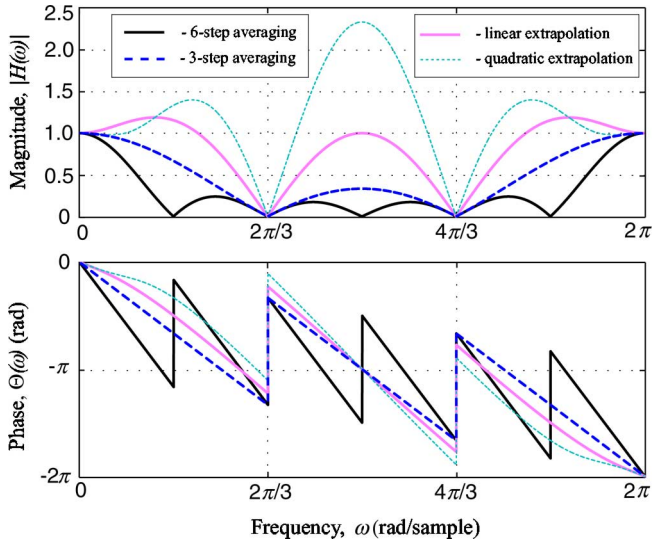


Fig. 11. Magnitude and phase responses of different filters.

After substituting (10) into (11), the resulting equation for computing $\bar{\tau}_l(n)$ in terms of $\tau(n)$ can be written as

$$\bar{\tau}_l(n) = \frac{1}{3}\left[2\tau(n-1) + \tau(n-2) + \tau(n-3) - \tau(n-4)\right] \tag{12}$$

which has the form of (6) and has fourth order. Fig. 9 shows the corresponding procedure for linear extrapolation and subsequent averaging to compute (12).

Higher order extrapolation is also possible. For example, the procedure of quadratic extrapolation and subsequent averaging is depicted in Fig. 10. Using this approach, the values $\tau_{ex_q}(n)$ are computed based on a three-step history and quadratic extrapolation as

$$\tau_{ex_q}(n) = 3\tau(n-1) - 3\tau(n-2) + \tau(n-3). \tag{13}$$

Then, the three values of $\tau_{ex_q}(n)$ are averaged as in (9), to obtain the following:

$$\bar{\tau}_q(n) = \frac{1}{3}\left[\tau_{ex_q}(n) + \tau_{ex_q}(n-1) + \tau_{ex_q}(n-2)\right]. \tag{14}$$

As with linear extrapolation, $\bar{\tau}_q(n)$ can be expressed in terms of $\tau(n)$ as

$$\bar{\tau}_q(n) = \frac{1}{3}\left[3\tau(n-1) + \tau(n-3) - 2\tau(n-4) + \tau(n-5)\right] \tag{15}$$

which also has the form of (6) and has fifth order.

C. Performance of Filters

To compare the performances of the proposed averaging filters, their magnitude and phase responses were calculated [9], [10]. The results are superimposed in Fig. 11. As can be observed, all of the filters completely rejected the undesirable harmonics with frequencies of $2\pi/3$ and $4\pi/3$ radians per sample, while perfectly retaining the dc component of the input signal. Therefore, all of these filters will achieve the required balancing of the modified Hall-sensor signals when the motor is in a steady state.

To compare the performances of the proposed averaging filters during speed transients, the filters were subjected to a linear acceleration assuming the same logic of the Hall sensors. In this test, a constant mechanical speed of 255 rad/s was initially applied to all of the filters. For the given 8-pole machine, this results in switching intervals $\tau(n)$ of approximately 1 ms, as depicted in Fig. 12. Then, at $t = 0.02$ s, the speed was linearly ramped with an acceleration of 13×10^3 rad/s^2 until it reached 320 rad/s at $t = 0.025$ s, after which the speed was kept constant. The transient responses produced by the considered filters are depicted in Fig. 12. To give the reader a better idea, the considered acceleration of 13×10^3 mechanical-rad/s^2 translates into 124.1×10^3 r/(min·s) (which is fairly high). To benchmark the filters, their performance was compared to the waveform of $\tau(n)$ produced by the Hall-sensor signals without any filter (ideal case, dashed line). As can be observed in Fig. 12, the response of various filters to the ramp test is noticeably different. The slowest response corresponds to the six-step moving-average filter (8), which is attributed to its longest memory. The successive improvement is demonstrated by the three-step filter (9) due to its shorter memory. At the same time, the filters based on linear and quadratic extrapolation [(12) and (15), respectively] both show very close transient responses, with the quadratic extrapolation filter demonstrating a slightly faster action at the beginning and end of the speed ramp.

IV. Reference Switching Time

Once the value $\bar{\tau}(n)$ is established using the appropriate filter, the actual timing for commutating the inverter transistors can be found as follows:

$$t_{\text{next_sw}} = \bar{t}(n) + \bar{\tau}(n) \tag{16}$$

where $\bar{t}(n)$ is the reference switching time and $\bar{\tau}(n)$ may denote $\bar{\tau}_{a6}(n)$, $\bar{\tau}_{a3}(n)$, $\bar{\tau}_l(n)$, or $\bar{\tau}_q(n)$. For example, this reference

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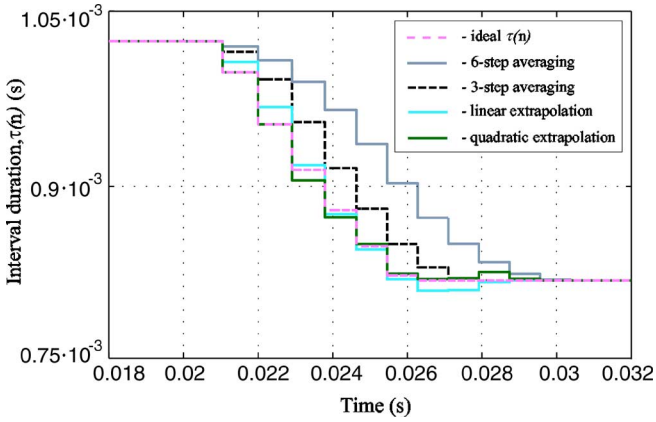


Fig. 12. Response of different filters to a ramp increase in speed.

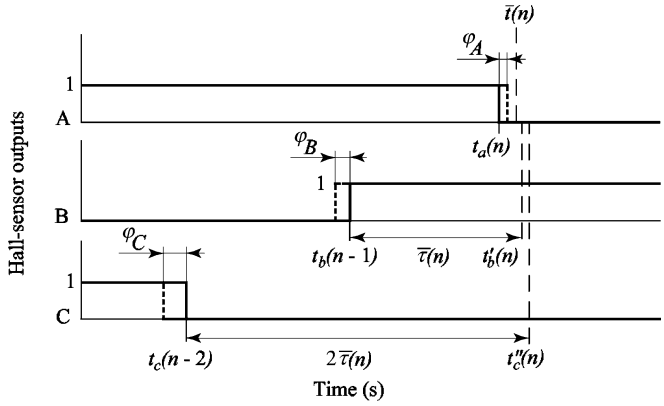


Fig. 13. Switching-event time relationships.

TABLE I
SWITCHING-EVENT TIME DETERMINATION

	$n-3$	$n-2$	$n-1$	n	$n+1$
A	$t_a(n-3)$	$t'_a(n-2)$	$t''_a(n-1)$	$t_a(n)$	$t'_a(n+1)$
B	$t'_b(n-3)$	$t''_b(n-2)$	$t_b(n-1)$	$t'_b(n)$	$t''_b(n+1)$
C	$t''_c(n-3)$	$t_c(n-2)$	$t'_c(n-1)$	$t''_c(n)$	$t_c(n+1)$
$\bar{\tau}$	$\bar{\tau}(n-3)$	$\bar{\tau}(n-2)$	$\bar{\tau}(n-1)$	$\bar{\tau}(n)$	$\bar{\tau}(n+1)$
$\bar{t}$	$\bar{t}(n-3)$	$\bar{t}(n-2)$	$\bar{t}(n-1)$	$\bar{t}(n)$	$\bar{t}(n+1)$
t_{next_sw}	$\bar{t}(n-4)+\bar{\tau}(n-4)$	$\bar{t}(n-3)+\bar{\tau}(n-3)$	$\bar{t}(n-2)+\bar{\tau}(n-2)$	$\bar{t}(n-1)+\bar{\tau}(n-1)$	$\bar{t}(n)+\bar{\tau}(n)$

time may be obtained by locking the switching to one of the phases (a phase with the smallest positioning error, if known) [9]. Alternatively, this time may be computed by averaging the switching times of the three phases [10], as follows:

$$\bar{t}(n) = \frac{1}{3}\left[t_*(n) + t'_*(n) + t''_*(n)\right] \tag{17}$$

where $t_*(n)$ is the time of the present switching phase and $t'_*(n)$ and $t''_*(n)$ are the times extrapolated from the other two phases, as follows:

$$\begin{aligned} t'_*(n) &= t_*(n-1) + \bar{\tau}(n) \\ t''_*(n) &= t_*(n-2) + 2\bar{\tau}(n). \end{aligned} \tag{18}$$

Here, the subscript "$*$" may denote phase A, B, or C, respectively.

For the purposes of illustration, the computation of the switching time estimates is summarized in Table I and shown in Fig. 13. Hence, if the most recent switching occurred in phase A, the reference time would be computed as

$$\bar{t}(n) = \frac{1}{3}\left[t_a(n) + t'_b(n) + t''_c(n)\right] \tag{19}$$

and thus, the $(n+1)$th switching in phase C would occur at $\bar{t}(n) + \bar{\tau}(n)$ instead of $t_c(n+1)$.

V. Implementation and Case Studies

In order to evaluate the performance of the proposed averaging filter, it was implemented in both the detailed model and the hardware prototype of the BLDC motor–inverter system. A programmable integrated circuit microcontroller PIC18F2331 [18] was used to allow flexibility in the filter implementation. A popular choice for motor drive applications, this microcontroller is often used for Hall-sensor-driven BLDC motors [19]. The filters proposed in (8), (9), (12), and (15) in conjunction with (16) were coded inside the section of the program that is triggered by a hardware interrupt coming from the Hall-sensor readings. This way, it is possible to perform all of the necessary filter calculations in a predictable amount of time (number of instructions) as well as determine the timings of firing the inverter transistors and schedule the corresponding interrupts.

Since all presented filters have memory, their usage imposes conditions on when the filters may be activated. For example, it is not possible to start a motor with the filter enabled, since, at the beginning, there is no previous history. Also, in the case of a very fast acceleration/deceleration transient, there potentially may be a need to deactivate the filter for some brief time, thereby defaulting to the existing Hall sensors after which the filter may be enabled again. A simplified block diagram of the motor controller allowing automatic enabling and disabling of the filter is shown in Fig. 14. Here, it is assumed that one of the proposed filters is used. To start the operation, the appropriate registers of the microcontroller have to be initialized. The variable "counter" counts the number of Hall-sensor transitions, whereas the "threshold" is set to the filter order plus one. After initialization, the controller checks the first IF condition. The purpose of this condition is to ensure that the filter is not used before its memory has stored sufficient data, and the motor starts using the original Hall-sensor signals for the first several switching transitions. After a sufficient number of transitions, the filter memory is ready, and the "counter" variable has been incremented to pass the first IF condition. For increased safety and reliability of the drive, the second IF condition checks to see if the motor is in any adverse transient by comparing the estimated acceleration/deceleration with some specified acceleration tolerance. If both the conditions are satisfied, the control of inverter transistors is performed using the modified (filtered) signals.

In the test implementation, the filter could also be enabled or disabled manually. To demonstrate the operation of the

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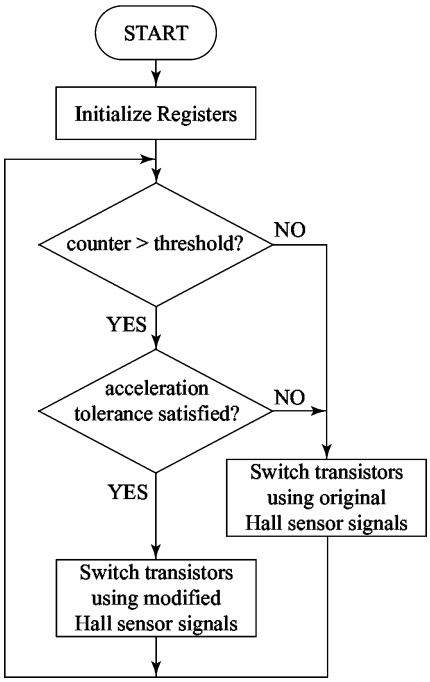


Fig. 14. High-level diagram of the microcontroller including the proposed filter.

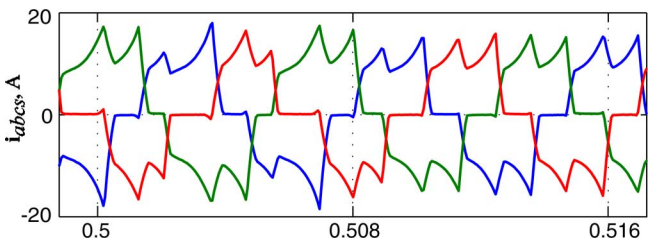


Fig. 15. Measured phase currents without and with the proposed filtering.

proposed filters in steady state, Fig. 15 shows a fragment of the measured stator currents corresponding to the dynamometer torque of about $1.4\ \mathrm{N \cdot m}$. Here, in the first part of the plot, the filter is disabled and the waveforms are clearly unbalanced, similar to those depicted in Fig. 4 (top). The filter is then enabled in the middle of Fig. 15, thereafter making the conduction intervals equal and the waveforms balanced among the phases, very similar to Fig. 4 (bottom).

A similar improvement of the phase currents was observed at different steady-state operating conditions from no-load to above the nominal load for every filter considered here.

A. Startup Transient

To illustrate the concept of automatic enabling of the filter, we ran experimental startup studies. To illustrate the performance of the motor in typical working conditions, the motor was mechanically coupled to a dynamometer with a combined inertia of $12 \times 10^{-4} \mathrm{N \cdot m \cdot s^2}$, while the inverter was supplied with 20 V dc to avoid overcurrent operation. For better comparisons among the filters, the initial position of the rotor was approximately aligned to the same reference position. The recorded transients are shown in Fig. 16. As can be seen, initially the motor operates with disabled filters producing very similar unbalanced currents. The filters are enabled at different times depending on the filter order, after which balanced operation among the motor phases is maintained. In each case, the motor accelerates following very similar speed trajectories, shown in Fig. 16 (bottom). In this study, the initial acceleration is around $4.76 \times 10^3\ \mathrm{rad/s^2}$ [$45.45 \times 10^3\ \mathrm{r/(\min \cdot s)}$]. However, by the time any filter is ready to be used, the acceleration decreases below $1.0 \times 10^3\ \mathrm{rad/s^2}$ [$9.55 \times 10^3\ \mathrm{r/(\min \cdot s)}$], at which point the difference among the responses of the filters becomes small too. Thereafter, as can be seen in Fig. 16, all filters demonstrate good transient performance achieving the desired balance among the phase currents. This study also demonstrates that the proposed filters do not compromise the startup performance of the drive.

B. Load-Step Transient

To investigate the dynamic performance of the BLDC motor with the proposed filters, we consider a transient caused by changing the dynamometer load. Since the dynamometer is a dc machine, the load change was implemented by changing the load resistor connected to its armature terminals. In this study, to achieve a rapid and appreciable change in speed, the effective load was increased from about 0.2 to 1.4 N·m, while the inverter was supplied with 40 V dc. Note that this large-signal transient spans the operating conditions of the drive beyond its ratings. The corresponding transient responses recorded without and with the proposed filtering are shown in Fig. 17. As Fig. 17 shows, when the filter is disabled (top subplot), the phase currents are unbalanced and spiky, similar to those depicted in Fig. 4 (top). In this study, all the previously described filters resulted in the same transient performance, achieving the desired balancing of the phase currents, as shown in Fig. 17 (middle). For the given total inertia of the system and the peak deceleration of $0.827 \times 10^3\ \mathrm{rad/s^2}$ [$7.9 \times 10^3\ \mathrm{r/(\min \cdot s)}$] as shown in Fig. 17 (bottom), even the slowest six-step filter performed adequately. This result is consistent with previous observations regarding the startup transient (see Fig. 16), wherein all filters performed well under acceleration of similar magnitude ($0.99 \times 10^3\ \mathrm{rad/s^2}$).

C. Voltage-Step Transient

To enable faster mechanical transients (similar to those considered in Fig. 12) and emulate the motor operation with small inertia, in the following studies, the dynamometer was disconnected, leaving the BLDC motor with a bare coupling. Initially, the machine was assumed to run in a steady state fed from 20 V dc with a total mechanical loss torque of about 0.1 N·m. At $t = 0.1$ s, the dc voltage was stepped up to 35 V dc, and the motor accelerated and continued to operate. Since in this test, direct measurement of speed and/or torque was not possible, both the detailed simulations and the hardware measurements were carried out.

The corresponding simulated speed and torque responses are shown in Figs. 18 and 19. For comparison, the transient of the BLDC drive system controlled without the filter is also given

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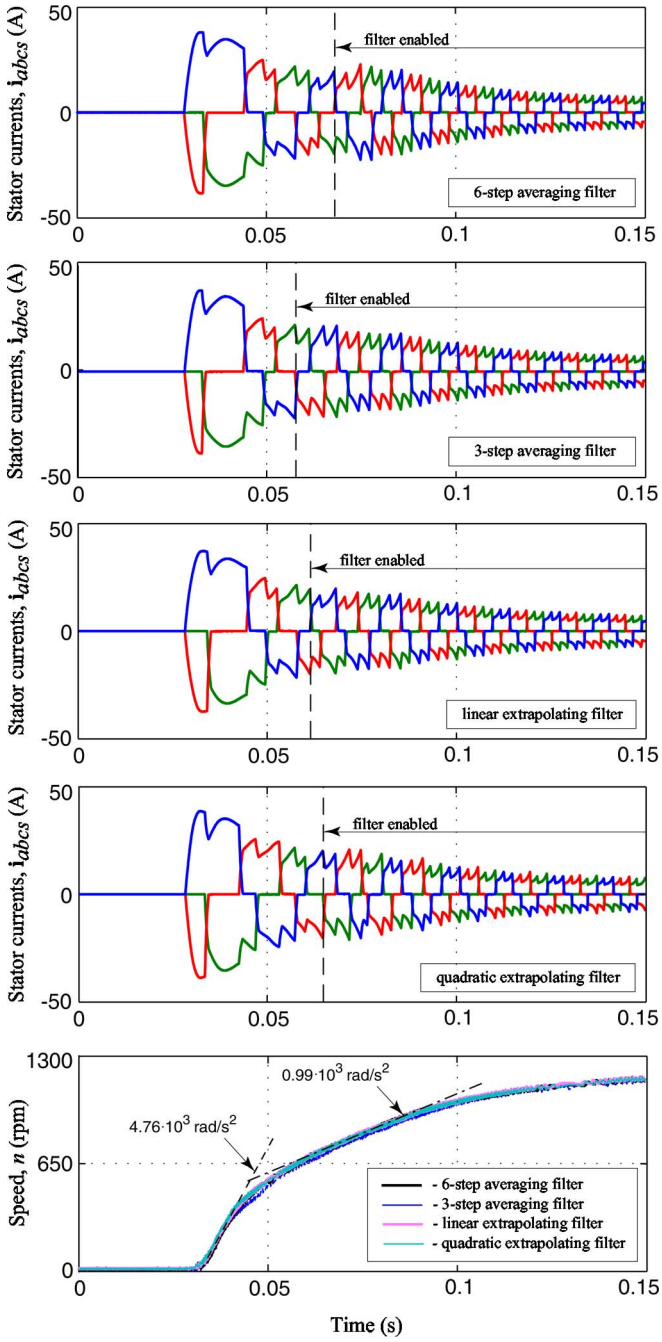


Fig. 16. Measured startup transient of BLDC motor.

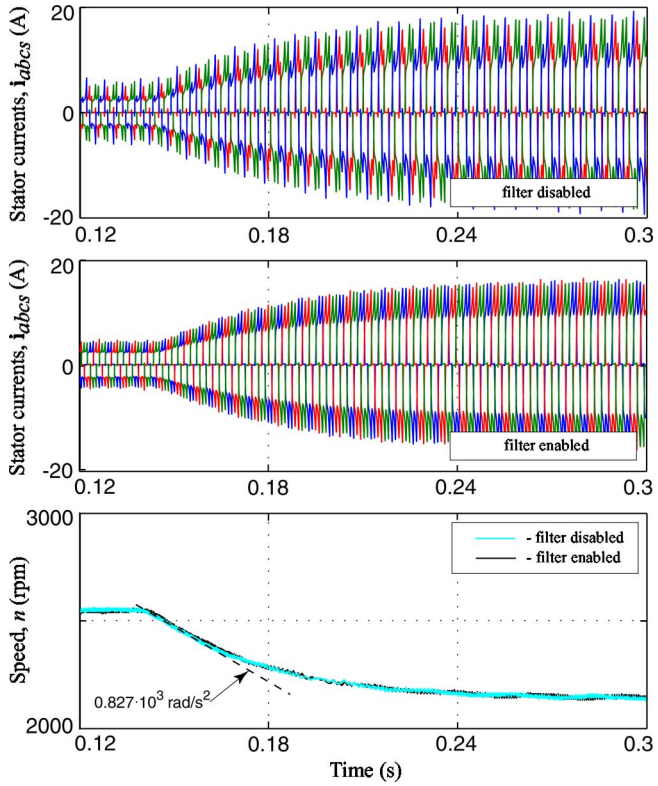


Fig. 17. Measured transient response due to load change.

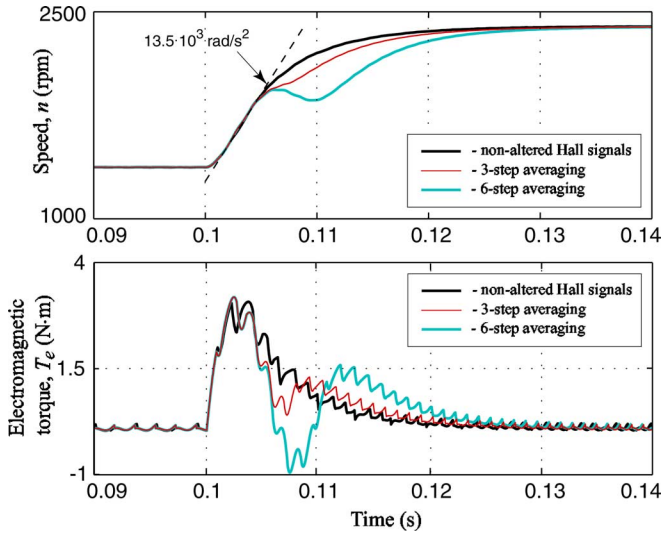


Fig. 18. Speed and electromagnetic torque response with three- and six-step averaging filters.

(black solid line). As can be seen, the increase in applied dc voltage was followed by a significant increase in developed electromagnetic torque and subsequent rapid acceleration of the motor. For this study, the peak acceleration was found to be 13.5×10^3 rad/s^2 [128.9×10^3 r/(min · s)]. This acceleration is by an order of magnitude higher than what was achieved in previous tests.

As was pointed out in Section III-C (see Fig. 12), the proposed filters will perform differently at very rapid changes of speed. The transients resulting from the six- and three-step averaging filters are compared in Fig. 18. As can be observed in Fig. 18, when either of the filters was used, the developed torque had a noticeable dip following several switching intervals, and then recovered. As expected, the three-step filter resulted in a smaller dip in torque and a faster recovery time than did the six-step filter, due to the difference in the memory capacities of these two filters. The corresponding delays are also noticed in the measured phase currents shown in Fig. 20 (first two subplots).

The transient responses produced by the BLDC motor with extrapolating averaging filters are shown in Fig. 19. As can

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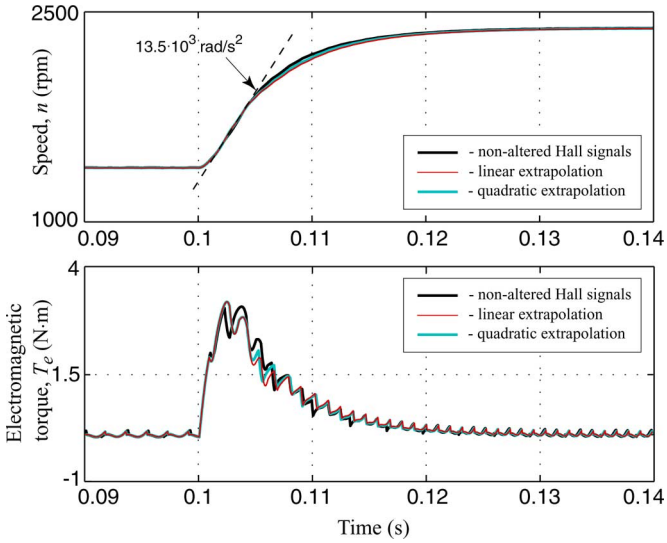


Fig. 19. Speed and electromagnetic torque response with extrapolating averaging filters.

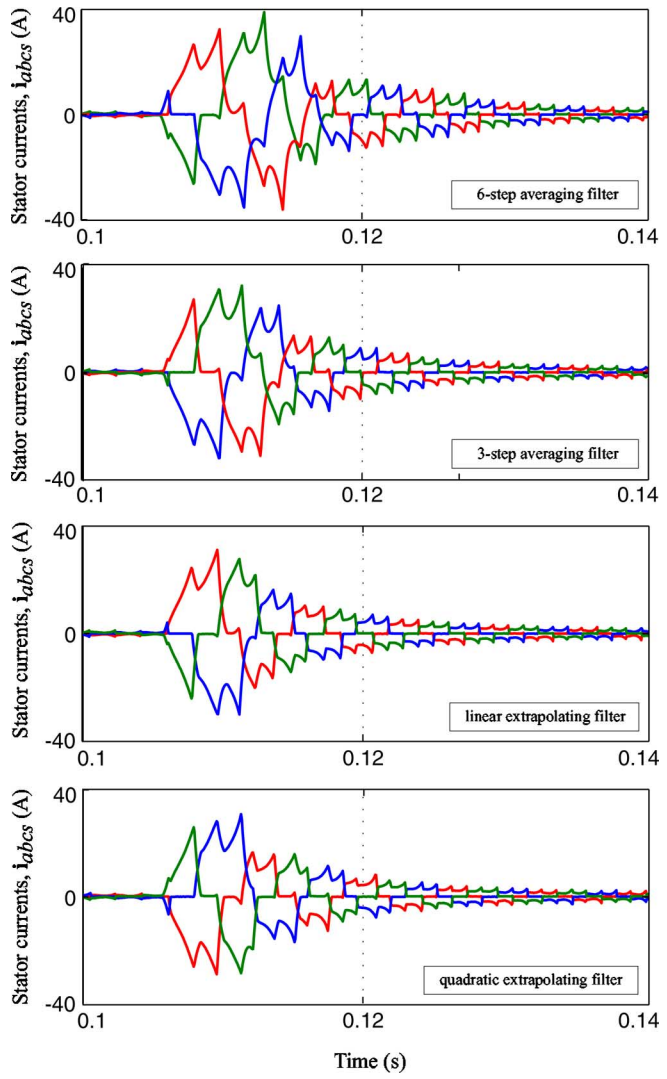


Fig. 20. Measured response of phase currents to step in dc voltage.

be seen, both extrapolating filters performed much faster than the basic moving-average filters, with almost no dip in torque and close to ideal speed response. The corresponding measured phase currents shown in Fig. 20 (third and fourth subplots) completely agree with this observation. This is an expected result, as the extrapolating filters where shown to cope very well with similar acceleration, as depicted in Fig. 12.

VI. Discussion

We have presented a number of studies that demonstrate the performance of BLDC motor drive with misaligned Hall sensors as well as with the proposed filtering approach. The studies were conducted in wide range of operating conditions including steady-state operation and transients below and above the ratings of the motor. As can be seen from the presented studies, the errors in Hall sensors cause unbalanced operation of the inverter leading to unequal conduction intervals among the motor phases. It is important to point out that this phenomenon does not depend on the loading and/or operating condition of the motor (low/high voltage, light/heavy load, etc.), and it will be present as long as the inverter transistors are commutated incorrectly. However, several observations can be made regarding the proposed filtering approach.

A. Steady-State Operation

It should be noted that all four filters described here resulted in absolutely the same steady-state performance, with complete balancing of the phase currents and rejection of the undesired low-frequency harmonics in torque, and therefore, performance approaching that of the ideally placed Hall sensors. However, due to the averaging of the original Hall-sensor signals, the corrected-balanced operation will correspond to the new firing advance angle

$$\varphi_v' = \varphi_v + \frac{(\varphi_A + \varphi_B + \varphi_C)}{3}. \tag{20}$$

This is a good result since the average of the absolute errors should be smaller than the largest individual error. In general, changes in firing advance angle φ_v affect the static torque–speed characteristic [1], Ch. [6], but small deviations should have minimal effect and the overall result should still be better than using the original unbalanced Hall sensors directly. Large deviations in φ_v may result in different operating modes. Interested reader will find definitions and analysis of several operating modes that may be obtained by varying φ_v in [3].

B. Transient Operation

As has been observed in the studies of Figs. 16 and 17, with larger mechanical inertia of the system (which results in a slower acceleration rate), the performance of all filters became very similar, with even the slowest six-step filter giving adequate transient performance. This approach, therefore, can be used in a large number of practical electromechanical and servo applications that commonly have significant effective inertia on the machine's shaft.

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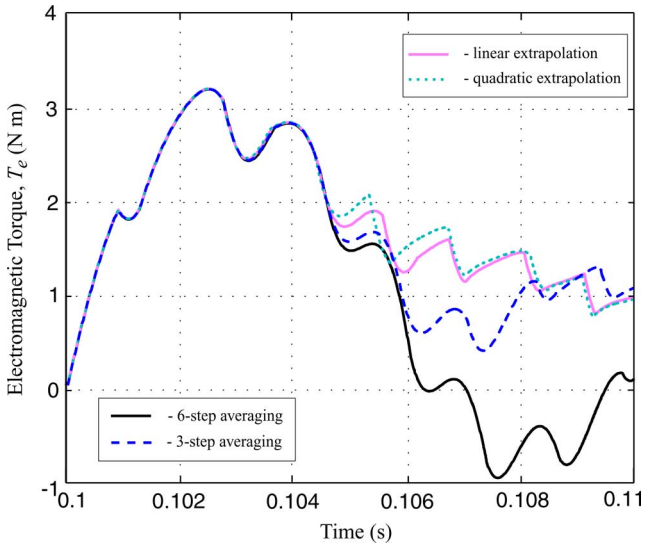


Fig. 21. Detailed view of electromagnetic torque response.

The transient study of Figs. 18–20 achieves high acceleration of 13.5×10^3 rad/s^2 [128.9×10^3 r/(min · s)] and clearly shows the differences among the proposed filters. During the first several switching intervals of the transient, due to the response delay of the basic six- and three-step averaging filters, the effective firing angle φ_v' also becomes momentarily delayed leading to a different operating mode. The change in operating mode can be observed in Fig. 20 (first two subplots), wherein the phase currents become continuous for just few of the switching intervals (six-step more pronounced and three-step less pronounced, respectively). This observation is consistent with analysis of operating modes presented in [3]. However, as can be seen in Fig. 20 (third and fourth subplots), this effect is much shorter for the extrapolating filters. Although it is difficult to compare the response of extrapolating filters in Fig. 20, a slightly different performance is observed in Fig. 19. For clarity, the predicted electromagnetic torque has been replotted again in Fig. 21 on a magnified scale. As Fig. 21 shows, during the first few switching intervals, the developed torque is the same for all cases. However, after $t = 0.105$ s, the results deviate showing that the quadratic extrapolation filter yields slightly higher torque and faster response among the considered cases. Moreover, both linear and quadratic extrapolating filters avoid the dip in torque that is present in the case of six- and three-step averaging filters, respectively. Therefore, for the systems with small inertia and/or very fast acceleration/deceleration requirements, the proposed extrapolating filters may offer a good solution.

VII. Conclusion

This paper presented a typical industrial low-precision BLDC motor and explained the phenomena of unbalanced Hall sensors. A detailed model of the considered motor drive has been developed and used to determine the effect of inaccurately placed Hall sensors on the resulting phase currents and developed electromagnetic torque. It was shown that unbalanced sensors lead to unequal conduction intervals among the motor phases and subsequently undesirable low-frequency harmonics in developed torque. Several filters have been proposed to improve steady-state and dynamic performance of such BLDC machine systems. Detailed simulations and hardware measurements were conducted to investigate the motor drive performance under wide range of operating conditions and support the analysis. A very good transient performance, approaching that of a motor with ideally placed Hall sensors, was achieved using the extrapolating and averaging filters applied to the signals from the original misaligned sensors.

Appendix

BLDC Machine Parameters: Arrow Precision Motor Corporation, Ltd., Model 86EMB3S98 F, 8 poles, $r_s = 0.14\,\Omega$, $L_s = 0.375$ mH, $\lambda_m' = 21$ mV·s; inertia $J = 2 \times 10^{-4}\mathrm{N \cdot m \cdot s^2}$; back EMF harmonic coefficients $K_1 = 1, K_3 = 0, K_5 = 0.042$, and $K_7 = -0.018$.

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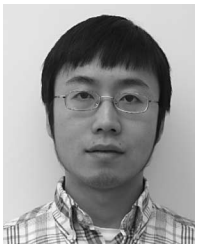
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